

E-COMMERCE COHORT CUSTOMER ANALYSIS 2024 -2025



Pipeline Logic & Data Flow (The Engine)

- **Source Connection:** The pipeline connects to the raw transaction database (**src_orders_2024_2025**) Notebook in databricks.
- **The Logic (Transformation Steps):**
 1. **First-Touch Identification:** The system scans all historical orders to find the absolute first purchase date for every unique Customer ID.
 2. **Cohort Assignment:** Customers are grouped into "Monthly Buckets" based on that first purchase date (e.g., all Feb 2024 starters stay in the "Feb 2024" group forever).
 3. **Behavioral Tracking:** The pipeline then cross-references these groups against all subsequent orders to calculate retention and repeat purchase depth.
- **Data Flow:**

Raw Data - Databricks Transformation - Cohort Metrics Table - Visualization Dashboard



Data Pipeline Architecture (The Madallion Framework)

- **Bronze Layer (Raw Data):**
 - **Logic:** Ingests the initial src_orders_2024_2025 data in its rawest form.
 - **Purpose:** Serves as a "Single Source of Truth." No transformations are applied yet, ensuring we can always re-process the data if logic changes.
- **Silver Layer (Cleaned & Augmented):**
 - **Logic:** Data is cleaned, filtered, deduplicated, and formatted before loading into the Gold-Layer
 - **Key Step:** This is where First-Purchase Dates, each order belongs to a specific customer by order_date, are calculated for each Customer ID to define the cohorts and Purchase Depth calculated here. This layer feeds the 2 charts shown in the report.
- **Gold Layer (Business Insights):**
 - **Logic:** Aggregated data specifically modeled for the Cohort Dashboard.
 - **Outcome:** Final metrics like retention_pct and cohort size calculated here. This layer feeds the 4 charts shown in the report.



Incremental Update & Analytical Impact

Goal: How the pipeline handled the 2025 update.

- **Handling the Update:**

- The **Bronze Layer** received the new 2025 transaction logs and appended these data to the existing Bronze-Layer
- The **Silver Layer** was cleaned, filtered, and formatted
- The **Gold Layer** recalculated the metrics, showing the "Quality over Quantity" shift in real-time.

- **The "Maturity" Filter:**

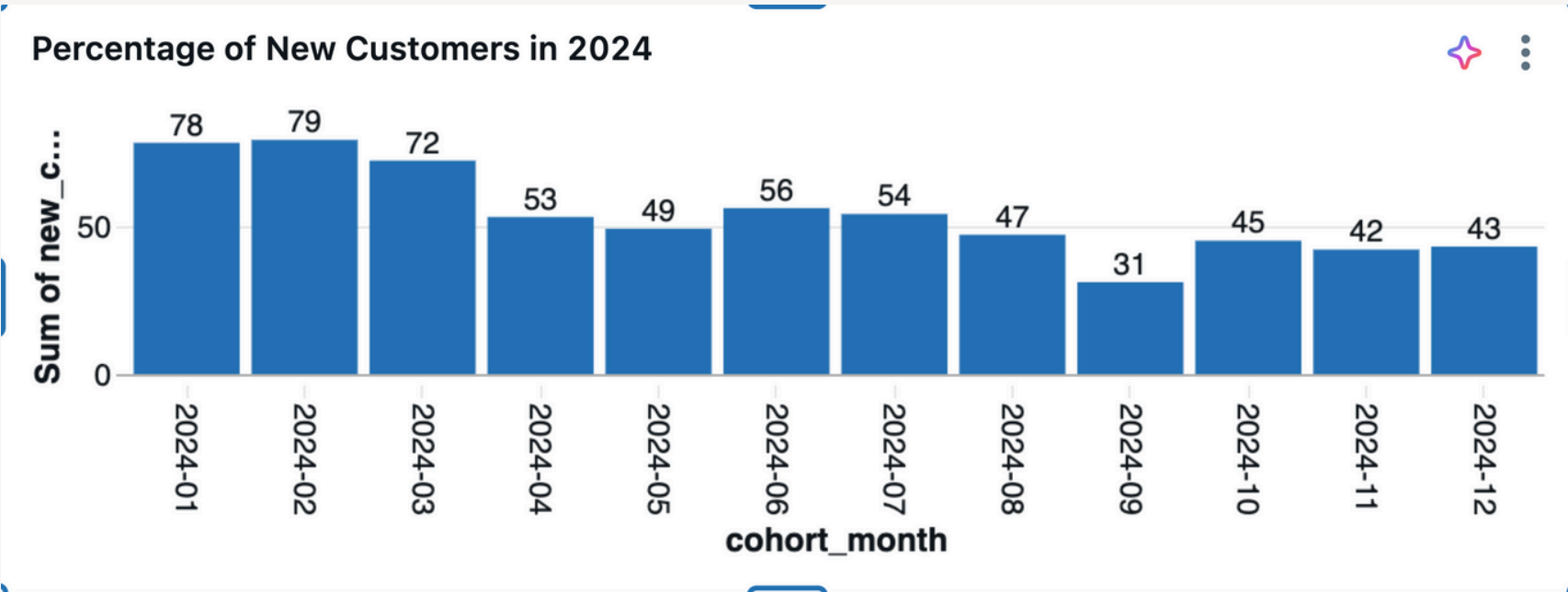
- The Gold Layer logic correctly reflects **Cohort Maturity**. Because 2025 cohorts are "younger," the pipeline naturally shows 0% for 4+ purchases—a deliberate design to reflect actual time-elapsed behavior.

The 2024 Baseline (Before Update)

TOTAL NEW CUSTOMERS (2024)

- **OBSERVATION:** HIGH VOLUME START (79 IN FEB) FOLLOWED BY A STEADY DECLINE.
- **STAGE CONCLUSION:** MARKETING REACH WAS NARROWING, BUT THE SCALE AS STILL SUFFICIENT.

WHAT WE KNEW BEFORE ADDING 2025 DATA

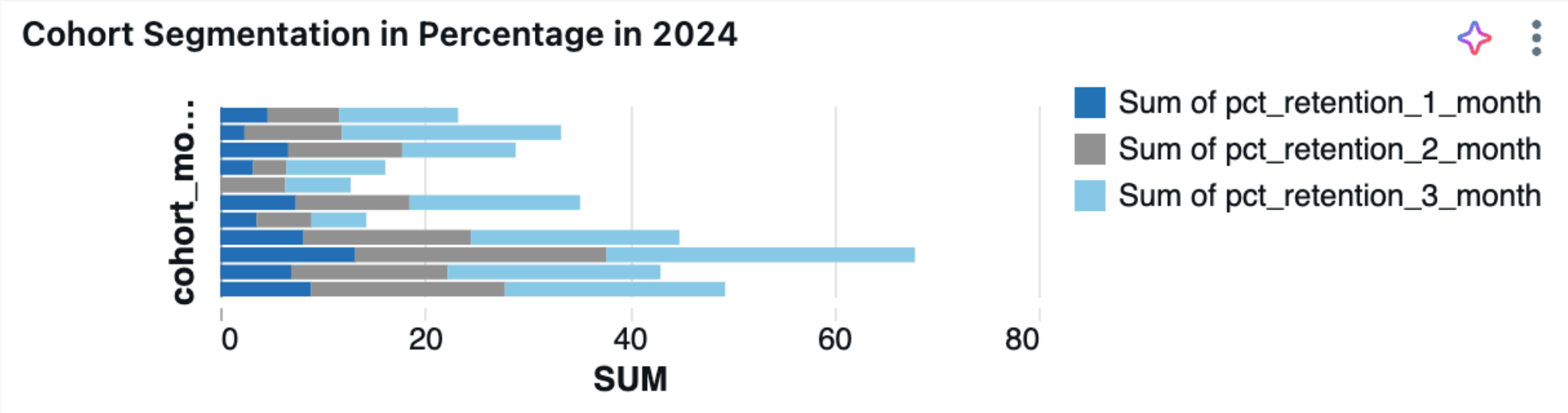


The 2024 Baseline (Before Update)

COHORT SEGMENTATION % (2024)

WHAT WE KNEW BEFORE ADDING 2025 DATA

- **OBSERVATION:** APRIL 2024 WAS THE "GOLDEN COHORT" WITH THE BEST 3-MONTH RETENTION.
- **STAGE CONCLUSION:** RETENTION WAS VOLATILE BUT PREDICTABLE WITHIN A 12-MONTH WINDOW.

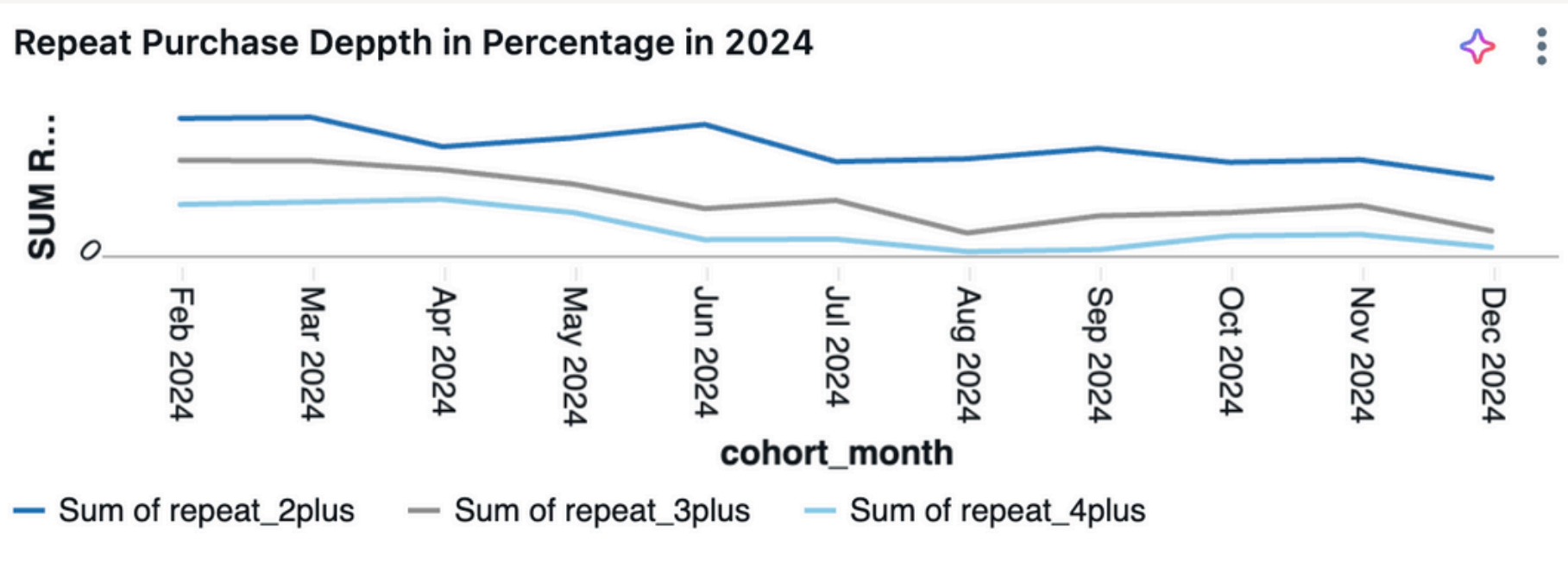


The 2024 Baseline (Before Update)

REPEAT PURCHASE DEPTH % (2024)

- **OBSERVATION:** NEARLY 50% OF CUSTOMERS MADE A 2ND PURCHASE IN EARLY COHORTS.
- **STAGE CONCLUSION:** WE HAD A STABLE "POWER USER" SEGMENT (4+ PURCHASES) OF 10–15%.

WHAT WE KNEW BEFORE ADDING 2025 DATA

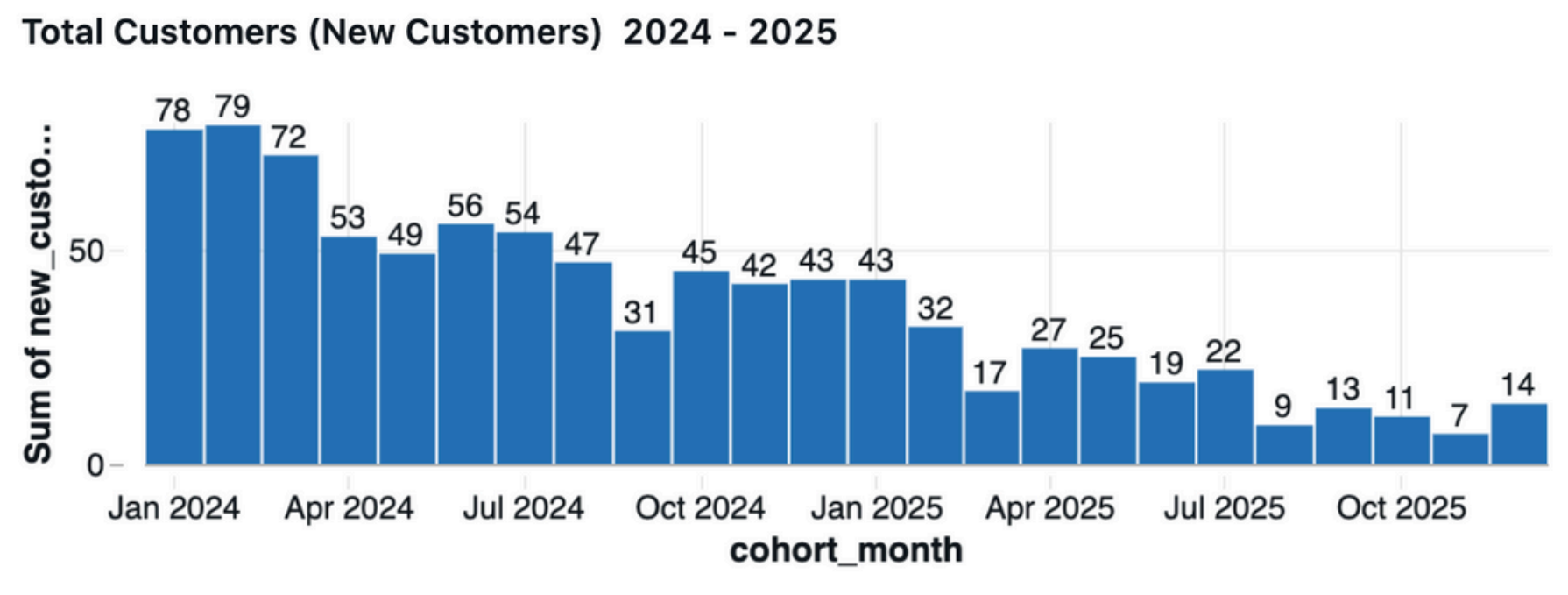


The Full 24-Month Picture (After Update)

TOTAL NEW CUSTOMER (2024-2025)

- **THE SHIFT:** THE 2024 DECLINE DIDN'T STOP; IT BOTTOMED OUT.
- **BUSINESS IMPACT:** BROAD ACQUISITION STRATEGIES ARE NO LONGER WORKING.

HOW 2025 DATA CHANGED OUR PERSPECTIVE

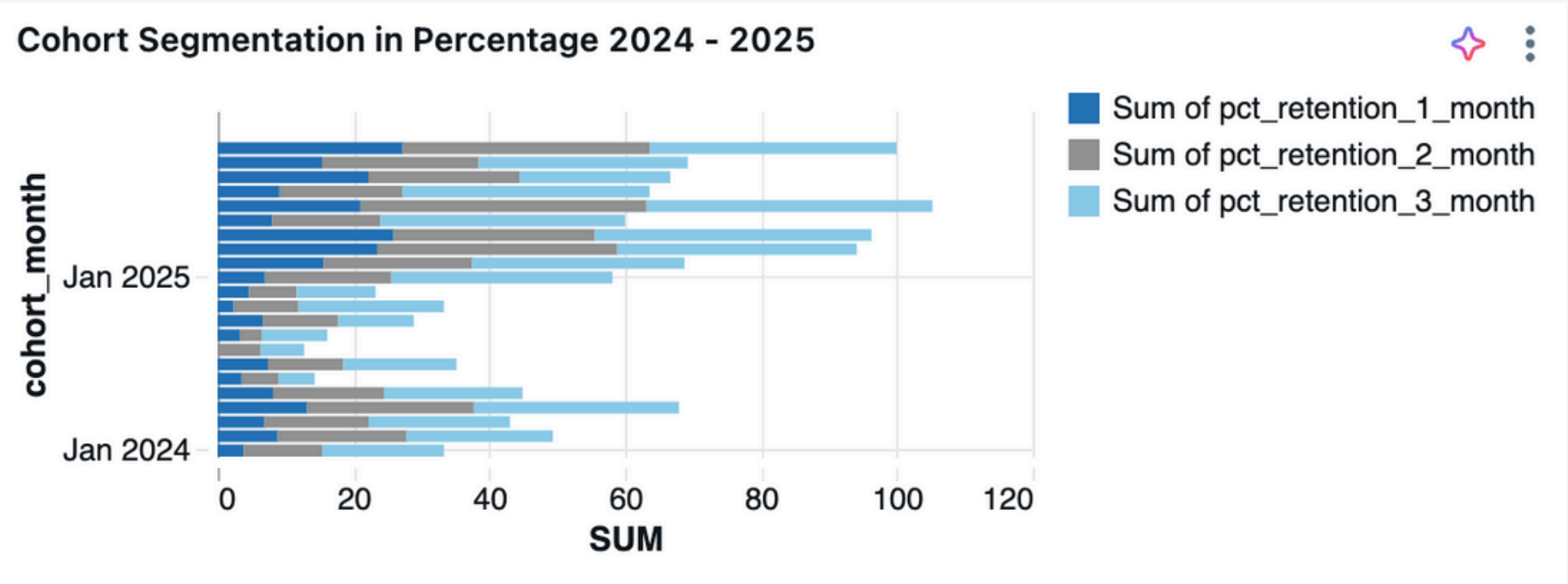


The Full 24-Month Picture (After Update)

COHORT SEGMENTATION % (2024-2025)

HOW 2025 DATA CHANGED OUR PERSPECTIVE

- **THE SHIFT:** RECENT COHORTS (JUNE/OCT 2025) SHOW RETENTION PEAKS DESPITE SMALL SIZES.
- **INSIGHT:** WE HAVE MOVED FROM A "MASS MARKET" MODEL TO A "NICHE/HIGH-LOYALTY" MODEL.

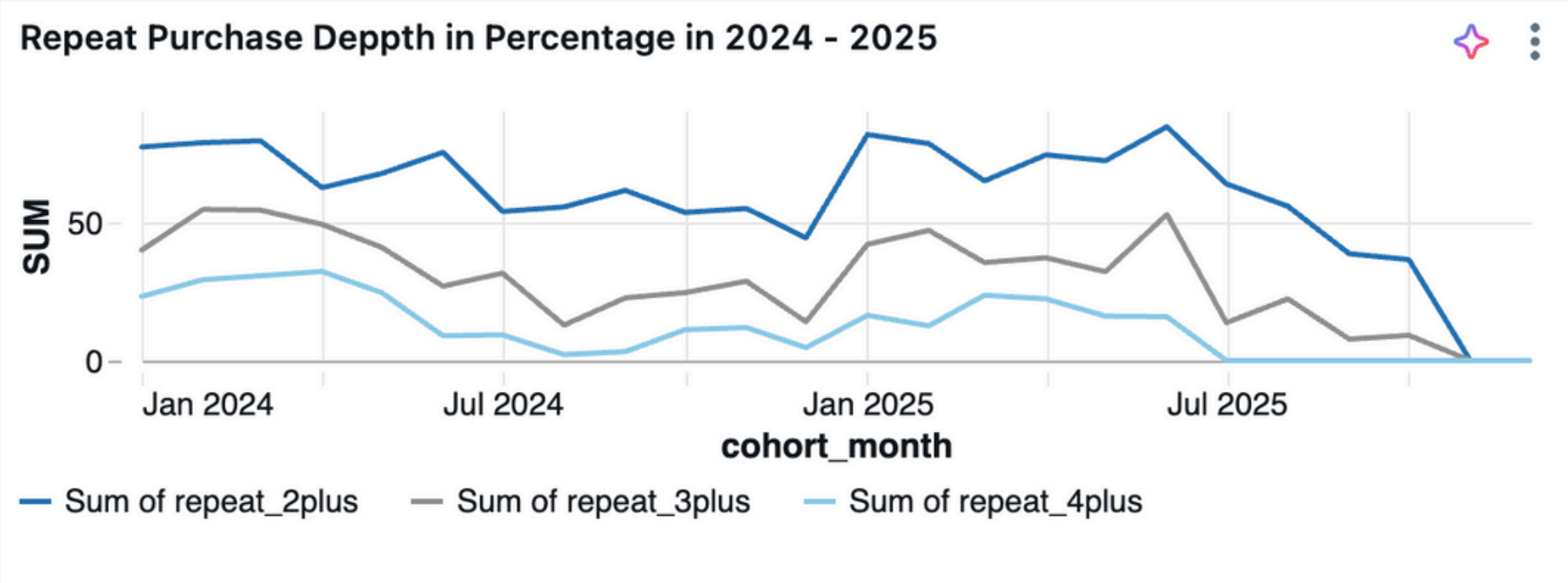


The Full 24-Month Picture (After Update)

REPEAT PURCHASE DEPTH % (2024-2025)

HOW 2025 DATA CHANGED OUR PERSPECTIVE

- **THE SHIFT:** AUGUST 2025 HIT A MASSIVE 55.56% 2ND PURCHASE RATE.
- **MATURITY NOTE:** THE DROP IN 4+ PURCHASES FOR LATE 2025 IS A "TIME-LAG" EFFECT, NOT A LOSS OF INTEREST.





Executive Summary

- **Acquisition Alert:** We are facing a sustained "Acquisition Crisis." New customer volume dropped from 79 (Feb 2024) to a critical low of 7 (Nov 2025).
- **Quality Paradox:** As volume decreased, customer quality increased. 2025 cohorts show higher relative retention than the 2024 average
- **Loyalty Benchmark:** Our core "2+ purchase" rate is healthy, often exceeding 50%
- **Data Integrity:**
 - The last day in 2024, 2024/12/31 is missing what effect the whole quality of this analysis.
 - The apparent 0% deep loyalty (3+ orders) in late 2025 is due to Cohort Maturity, not failure—these customers simply need more time



Key Results (the 6 Data Pillars)

1. Acquisition Trends (Charts 1 & 4)

- **Key Result:** A 91% drop in peak acquisition volume (from 79 in Feb 2024 to 7 in Nov 2025).
- **Conclusion:** The acquisition engine has reached a critical failure point.

2. Retention Performance (Charts 2 & 5)

- **Key Result:** 2025 cohorts (specifically June and October) outperformed 2024 cohorts in 1-3 month retention percentages.
- **Conclusion:** Recent marketing/product changes are attracting stickier customers despite lower traffic.

3. Loyalty Depth (Charts 3 & 6)

- **Key Result:** Core loyalty (2+ purchases) peaked in August 2025 at 55.56%.
- **Conclusion:** Once a customer is acquired, their likelihood of returning is higher now than it was in 2024.

4. Cohort Maturity Lag

- **Key Result:** 3+ and 4+ purchase metrics show 0% for late 2025 cohorts.
- **Conclusion:** This is a confirmed "time-lag" effect. These cohorts are too young to show deep loyalty results yet, making them "dark horse" opportunities for 2026 revenue.



Strategic Recommendations

1. Audit Acquisition vs. Quality

- **Action:** Analyze if the 2025 "High Quality" shift is intentional. If we are spending the same budget for 7 customers as we did for 79, the ROI is failing despite the high loyalty.

2. Launch Single-Purchase Re-activation and missing data from 2024-12-31

- **Action:** Isolate "One-Time Buyers" from high-volume 2024 cohorts and collect the data from the 2024-12-31
- **Target:** Re-engaging just 10% of the 2024 "lost" customers would be more impactful than current 2025 acquisition and Data completion regarding missing Silvester-day in 2024

3. Support the "Power User" Core

- **Action:** Since 2025 customers are inherently more loyal, implement a VIP tier specifically to protect the 4+ purchase segment.

4. Maturity-Based Forecasting

- **Action:** Do not cut budgets based on Q4 2025 "0% loyalty" metrics. Re-evaluate these in July 2026 to see their true lifetime value.



THANK YOU

Please get in touch if you have any questions or concerns about this report.